

Skew monoidality

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- ▶ **Quantum categories and bialgebroids**
- ▶ **Coherence for skew monoidal categories**

Skew monoidal categories

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- ▶ natural families of (not necessarily invertible) morphisms

$$\alpha : A \otimes (B \otimes C) \longrightarrow (A \otimes B) \otimes C$$

$$\lambda : A \longrightarrow I \otimes A$$

$$\rho : A \otimes I \longrightarrow A$$

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- ▶ plus five axioms as follows:

The pentagon and trivial unit condition

$$\begin{array}{ccc}
 & (W \otimes X) \otimes (Y \otimes Z) & \\
 \nearrow \alpha_{W,X,Y \otimes Z} & & \searrow \alpha_{W \otimes X,Y,Z} \\
 W \otimes (X \otimes (Y \otimes Z)) & & ((W \otimes X) \otimes Y) \otimes Z \\
 \downarrow 1_W \otimes \alpha_{X,Y,Z} & & \uparrow \alpha_{W,X,Y} \otimes 1_Z \\
 W \otimes ((X \otimes Y) \otimes Z) & \xrightarrow{\alpha_{W,X \otimes Y,Z}} & (W \otimes (X \otimes Y)) \otimes Z
 \end{array}
 \tag{1}$$

$$\begin{array}{ccc}
 I & \xrightarrow{1_I} & I \\
 \searrow \lambda_I & & \nearrow \rho_I \\
 & I \otimes I &
 \end{array}
 \tag{2}$$

Three more unit conditions

$$\begin{array}{ccc}
 I \otimes (X \otimes Y) & \xrightarrow{\alpha_{I,X,Y}} & (I \otimes X) \otimes Y \\
 & \nwarrow \lambda_{X \otimes Y} \quad \nearrow \lambda_X \otimes 1_Y & \\
 & X \otimes Y &
 \end{array} \tag{3}$$

$$\begin{array}{ccc}
 X \otimes (I \otimes Y) & \xrightarrow{\alpha_{X,I,Y}} & (X \otimes I) \otimes Y \\
 1_X \otimes \lambda_Y \uparrow & & \downarrow \rho_X \otimes 1_Y \\
 X \otimes Y & \xrightarrow{1_{X \otimes Y}} & X \otimes Y
 \end{array} \tag{4}$$

$$\begin{array}{ccc}
 X \otimes (Y \otimes I) & \xrightarrow{\alpha_{X,Y,I}} & (X \otimes Y) \otimes I \\
 & \searrow 1_X \otimes \rho_Y \quad \swarrow \rho_{X \otimes Y} & \\
 & X \otimes Y &
 \end{array} \tag{5}$$

Skew monoidales

A *right skew-monoidal* structure on an object R in a monoidal bicategory $\mathcal{M}$ consists of morphisms $p : R \otimes R \rightarrow R$, $j : I \rightarrow R$, respectively called the *tensor product* and *unit*, and 2-cells

$$\begin{array}{ccc}
 R \otimes R \otimes R & \xrightarrow{p \otimes 1} & R \otimes R \\
 \downarrow 1 \otimes p & \xRightarrow{\alpha} & \downarrow p \\
 R \otimes R & \xrightarrow{p} & R
 \end{array}$$

$$\begin{array}{ccc}
 R \otimes R & \xleftarrow{j \otimes 1} & R \\
 \searrow p & \xleftarrow{\lambda} & \swarrow 1 \\
 & R &
 \end{array}
 \qquad
 \begin{array}{ccc}
 R \otimes R & \xleftarrow{1 \otimes j} & R \\
 \searrow p & \xleftarrow{\rho} & \swarrow 1 \\
 & R &
 \end{array}$$

respectively called the *associativity*, *left unit* and *right unit constraints*, satisfying five conditions expressible using pasting diagrams (Exercise).

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- ▶ Objects A of $\mathbf{Comon}\mathcal{V}\text{-Cat} = (\mathbf{Comon}\mathcal{V})\text{-Cat}$ might be considered locally non-commutative $\mathcal{V}$ -categories. Put $X = \text{ob}A$ assumed small.

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$$(M *_A N)_x = \sum_y M_x \otimes A(x, y) \otimes N_y .$$

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- ▶ The unit object J is defined by $J_x = I$ for all $x \in X$.

Example of a skew monoidal category (continued)

$$(M *_A (N *_A L))_x = \sum_{y,z} M_x \otimes A(x, y) \otimes N_y \otimes A(y, z) \otimes L_z$$

$$((M *_A N) *_A L)_x = \sum_{y,z} M_x \otimes A(x, y) \otimes N_y \otimes A(x, z) \otimes L_z$$

- The morphism $\alpha : M *_A (N *_A L) \longrightarrow (M *_A N) *_A L$ uses

$$A(x, y) \otimes A(y, z) \xrightarrow{\delta \otimes 1} A(x, y) \otimes A(x, y) \otimes A(y, z) \xrightarrow{1 \otimes \text{comp}} A(x, y) \otimes A(x, z).$$

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- ▶ The morphism $\lambda : N \longrightarrow J *_A N$ uses

$$N_x \xrightarrow{\text{id}_x \otimes 1} A(x, x) \otimes N_x \xrightarrow{\text{in}_x} \sum_y A(x, y) \otimes N_x.$$

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- ▶ The morphism $\rho : M *_A J \longrightarrow M$ uses $\sum_y M_x \otimes A(x, y) \xrightarrow{1 \otimes \epsilon} M_x$.

Opmonoidal monads

An opmonoidal monad $t = (t : R \rightarrow R, \mu : tt \Rightarrow t, \eta 1 \Rightarrow t)$ on a right skew monoidal R has 2-cells

$$\begin{array}{ccc}
 R \otimes R & \xrightarrow{t \otimes t} & R \otimes R \\
 p \downarrow & \xRightarrow{\delta} & \downarrow p \\
 R & \xrightarrow{t} & R
 \end{array}
 \qquad
 \begin{array}{ccc}
 I & \xrightarrow{j} & R \\
 j \searrow & \xleftarrow{\epsilon} & \nearrow t \\
 & R &
 \end{array}$$

making t , μ and η opmonoidal. The fusion 2-cell v is

$$\begin{array}{ccc}
 R \otimes R & \xrightarrow{t \otimes 1} & R \otimes R \\
 1 \otimes t \downarrow & \xRightarrow{1_t \otimes \mu} & \downarrow 1 \otimes t \\
 R \otimes R & \xrightarrow{t \otimes t} & R \otimes R \\
 p \downarrow & \xRightarrow{\delta} & \downarrow p \\
 R & \xrightarrow{t} & R
 \end{array}
 \tag{6}$$

Fusion structures

Bruguères-Lack-Virelizier (2011) define the opmonoidal monad to be *Hopf* when v is invertible.

Theorem

Opmonoidal monad structures $(\mu, \eta, v, \varepsilon)$ on an endomorphism t of a right skew monoidal R are in bijection, via the definition (6) of v , with triples (v, η, ε) of 2-cells satisfying five specific axioms.

Here are the five axioms.

$$\begin{array}{ccc} p(1 \otimes t) & \xrightarrow{\eta p(1 \otimes t)} & tp(1 \otimes t) \\ & \searrow p(\eta \otimes 1) \quad \swarrow v & \\ & p(t \otimes t) & \end{array}$$

$$\begin{array}{ccc} j & \xrightarrow{\eta j} & tj \\ & \searrow 1_j \quad \swarrow \varepsilon & \\ & j & \end{array}$$

$$\begin{array}{ccccc}
 & & p(t \otimes t)(1 \otimes j) & & \\
 & \nearrow^{v(1 \otimes j)} & & \searrow^{\cong} & \\
 tp(1 \otimes t)(1 \otimes t) & & & & p(1 \otimes t)(1 \otimes j)t \\
 \downarrow tp(1 \otimes \varepsilon) & & & & \downarrow \rho(1 \otimes \varepsilon)t \\
 tp(1 \otimes j) & \xrightarrow{\rho} & t & \xleftarrow{\rho} & p(1 \otimes j)t
 \end{array}$$

$$\begin{array}{ccc}
tp(1 \otimes t)(j \otimes 1) & \xrightarrow{v(j \otimes 1)} & p(t \otimes t)(j \otimes 1) \\
\uparrow tp(1 \otimes \eta)(j \otimes 1) & & \downarrow \cong \\
tp(j \otimes 1) & & p(tj \otimes 1)t \\
\uparrow t\lambda & & \downarrow p(\varepsilon \otimes 1)t \\
t & \xrightarrow{\lambda t} & p(j \otimes 1)t
\end{array}$$

$$\begin{array}{ccc}
tp(1 \otimes t)(1 \otimes p)(1 \otimes 1 \otimes t) & \xrightarrow{v(1 \otimes p)(1 \otimes 1 \otimes t)} & p(t \otimes t)(1 \otimes p)(1 \otimes 1 \otimes t) \\
\downarrow tp(1 \otimes v) & & \downarrow p(t \otimes 1)(1 \otimes v) \\
tp(1 \otimes p)(1 \otimes t \otimes t) & & p(t \otimes 1)(1 \otimes p)(1 \otimes t \otimes t) \\
\downarrow \alpha & & \downarrow \alpha \\
tp(1 \otimes t)(p \otimes 1)(1 \otimes t \otimes 1) & & p(p \otimes 1)(t \otimes t \otimes t) \\
\downarrow v(p \otimes 1)(1 \otimes t \otimes 1) & & \uparrow \cong \\
p(t \otimes t)(p \otimes 1)(1 \otimes t \otimes 1) & \xrightarrow{p(1 \otimes t)(v \otimes 1)} & p(1 \otimes t)(p \otimes 1)(t \otimes t \otimes 1)
\end{array}$$

Skew flocks

A *skew flock* in a monoidal bicategory $\mathcal{M}$ is an object R equipped with a right bidual R° , having unit and counit $n : I \longrightarrow R^\circ \otimes R$ and $e : R \otimes R^\circ \longrightarrow I$, a morphism

$$q : R \otimes R^\circ \otimes R \longrightarrow R$$

and 2-cells

$$\begin{array}{ccc}
 R \otimes R^\circ \otimes R \otimes R^\circ \otimes R & \xrightarrow{q \otimes 1 \otimes 1} & R \otimes R^\circ \otimes R \\
 \downarrow 1 \otimes 1 \otimes q & \uparrow \phi & \downarrow q \\
 R \otimes R^\circ \otimes R & \xrightarrow{q} & R
 \end{array} \tag{7}$$

$$\begin{array}{ccc}
 & R \otimes R^\circ \otimes R & \\
 1 \otimes n \nearrow & \Downarrow \zeta & \searrow q \\
 R & \xrightarrow{\quad 1 \quad} & R
 \end{array}
 \tag{8}$$

$$\begin{array}{ccc}
 & e \otimes 1 & \\
 R \otimes R^\circ \otimes R & \begin{array}{c} \xrightarrow{\quad} \\ \Downarrow \zeta \\ \xrightarrow{\quad} \end{array} & R \\
 & q &
 \end{array}
 \tag{9}$$

satisfying the following five conditions (where $1_n = \overbrace{1 \otimes \dots \otimes 1}^n$):

$$\begin{array}{ccc}
 & q(1_2 \otimes q)(q \otimes 1_4) \cong q(q \otimes 1_2)(1_4 \otimes q) & \\
 \swarrow \phi(q \otimes 1_4) & & \nwarrow \phi(1_4 \otimes q) \\
 (q(q \otimes 1_2)(q \otimes 1_4)) & & q(1_2 \otimes q)(1_4 \otimes q) \\
 \uparrow q(\phi \otimes 1_2) & & \downarrow q(1_2 \otimes \phi) \\
 q(q \otimes 1_2)(1_2 \otimes q \otimes 1_2) & \xleftarrow{\phi(1_2 \otimes q \otimes 1_2)} & q(1_2 \otimes q)(1_2 \otimes q \otimes 1_2)
 \end{array}$$

$$\begin{array}{ccc}
 1 & \cong & (e \otimes 1)(1 \otimes n) \xrightarrow{\zeta(1 \otimes n)} q(1 \otimes n) \\
 & \searrow & \downarrow \xi \\
 & & 1
 \end{array}$$

1

$$\begin{array}{ccc}
 (e \otimes 1)(1_2 \otimes q) & \xrightarrow{\zeta(1_2 \otimes q)} & q(1_2 \otimes q) \\
 \downarrow \cong & & \downarrow \phi \\
 q(e \otimes 1_3) & \xrightarrow{q(\zeta \otimes 1_2)} & q(q \otimes 1_2)
 \end{array}$$

$$\begin{array}{ccc}
 q(1_2 \otimes q)(1_3 \otimes n) & \xrightarrow{\phi(1_3 \otimes n)} & q(q \otimes 1_2)(1_3 \otimes n) \\
 \downarrow q(1_2 \otimes \xi) & & \downarrow \cong \\
 q & \xleftarrow{\xi q} & q(1 \otimes n)q
 \end{array}$$

$$\begin{array}{ccc}
q(1_2 \otimes q)(1_1 \otimes n \otimes 1_2) & \xrightarrow{\phi(1_1 \otimes n \otimes 1_2)} & q(q \otimes 1_2)(1_1 \otimes n \otimes 1_2) \\
\uparrow q(1_2 \otimes \zeta)(1_1 \otimes n \otimes 1_2) & & \downarrow q(\xi \otimes 1_2) \\
q(1_2 \otimes e \otimes 1_1)(1_1 \otimes n \otimes 1_2) \cong q1_3 \cong q & \xrightarrow{1} & q \cong q1_3
\end{array}$$

A *flock* as per Booker-Street (2011) is a skew flock with ϕ invertible.

Example of a skew flock

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- ▶ Put $\text{ev} = \pi^{-1}(1_{\mathcal{A}(B, C)}) : \mathcal{A}(B, C) \star B \longrightarrow C$ and $\text{id} = \pi(I \star B \cong B) : I \longrightarrow \mathcal{A}(B, B)$.

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- ▶ Let $Q : \mathcal{A} \otimes \mathcal{A}^{\text{op}} \otimes \mathcal{A} \longrightarrow \mathcal{A}$ be the functor

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$$Q(A, B, C) = \mathcal{A}(B, A) \star C .$$

- ▶ $\pi((\mathcal{A}(B, A) \otimes \mathcal{A}(D, C)) \star D \cong \mathcal{A}(B, A) \star (\mathcal{A}(D, C) \star D) \xrightarrow{1 \otimes \text{ev}} \mathcal{A}(B, A) \star C)$

$$= (\mathcal{A}(B, A) \otimes \mathcal{A}(D, C) \longrightarrow \mathcal{A}(D, \mathcal{A}(B, A) \star C))$$

to which we apply $- \star E$ to obtain a natural family

$$\phi : Q(A, B, Q(C, D, E)) \longrightarrow Q(Q(A, B, C), D, E) .$$

- ▶ The evaluation $\text{ev} : \mathcal{A}(A, B) \star A \longrightarrow A$ gives a natural family

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- ▶ $\text{id} \otimes 1 : A \longrightarrow \mathcal{A}(B, B) \star A$ gives a natural family

$$\zeta : A \longrightarrow Q(B, B, A) .$$

This defines a skew flock $(\mathcal{A}, Q_*, \phi, \xi, \zeta)$ in the putative monoidal bicategory $\mathcal{V}\text{-Mod}$.

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of $q : R \otimes R^\circ \otimes R \longrightarrow R$ under the biduality $R \dashv_b R^\circ$.

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Proposition

There is a bijection between skew flock structures on (R, q) and opmonoidal monad structures on $(R^\circ \otimes R, \hat{q})$.

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- ▶ Let $j : I \longrightarrow R$ have a right adjoint j^* with counit $\varepsilon_j : j \circ j^* \Longrightarrow 1_R$.
- ▶ Put $h = j^{*\circ} : I \longrightarrow R^\circ$. Then h also has a right adjoint $h^* = j^\circ$.

Moreover, $j^* = e \circ (1 \otimes h)$, $h \cong (1 \otimes j^*) \circ n$ and

$j \circ j^* \cong (j^* \otimes 1_R) \circ (1_R \otimes j) \cong j^* \otimes j$. So $(h \otimes 1) \circ j \cong j \circ j^* \circ n$.

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- ▶ Define

$$p = (R \otimes R \xrightarrow{1 \otimes h \otimes 1} R \otimes R^\circ \otimes R \xrightarrow{q} R) .$$

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- ▶ Put $h = j^{*\circ} : I \longrightarrow R^\circ$. Then h also has a right adjoint $h^* = j^\circ$.
Moreover, $j^* = e \circ (1 \otimes h)$, $h \cong (1 \otimes j^*) \circ n$ and
 $j \circ j^* \cong (j^* \otimes 1_R) \circ (1_R \otimes j) \cong j^* \otimes j$. So $(h \otimes 1) \circ j \cong j \circ j^* \circ n$.
- ▶ Define

$$p = (R \otimes R \xrightarrow{1 \otimes h \otimes 1} R \otimes R^\circ \otimes R \xrightarrow{q} R) .$$

- ▶ Obtain $\alpha : p(1 \otimes p) \Longrightarrow p(p \otimes 1)$ from $\phi(1 \otimes h \otimes 1 \otimes h \otimes 1) :$
 $q(1 \otimes 1 \otimes q)(1 \otimes h \otimes 1 \otimes h \otimes 1) \Longrightarrow q(q \otimes 1 \otimes 1)(1 \otimes h \otimes 1 \otimes h \otimes 1)$
and the canonical isomorphisms

$$p(1 \otimes p) \cong q(1 \otimes 1 \otimes q)(1 \otimes h \otimes 1 \otimes h \otimes 1)$$

and

$$q(q \otimes 1 \otimes 1)(1 \otimes h \otimes 1 \otimes h \otimes 1) \cong p(p \otimes 1) .$$

- ▶ Next $\lambda : 1_R \Rightarrow p \circ (j \otimes 1_R)$ has mate $j^* \otimes 1 \Rightarrow p$ under the adjunction $j \otimes 1 \dashv j^* \otimes 1$ equal to

$$\zeta \circ (1 \otimes h \otimes 1) : (e \otimes 1) \circ (1 \otimes h \otimes 1) \Rightarrow q \circ (1 \otimes h \otimes 1) .$$

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where the unlabelled 2-cell is $q \circ (1 \otimes \varepsilon_j) \circ (1 \otimes n)$.

Proposition

For any skew flock R and any left adjoint morphism $j : I \longrightarrow R$ in $\mathcal{M}$, with p, α, λ and ρ as above, (R, p, j) is a right skew monoidale in $\mathcal{M}$.

Definition

A *right skew $\mathcal{V}$ -proflock* is a right skew flock $\mathcal{A}$ in $\mathcal{V}\text{-Mod}$.

A *right skew $\mathcal{V}$ -flock* is a right skew $\mathcal{V}$ -proflock $\mathcal{A}$ in $\mathcal{V}\text{-Mod}$ for which the module

$$Q : \mathcal{A} \otimes \mathcal{A}^{\text{op}} \otimes \mathcal{A} \longrightarrow \mathcal{A} \quad (10)$$

is (representable as) a $\mathcal{V}$ -functor. More explicitly, we have a $\mathcal{V}$ -category $\mathcal{A}$ and a $\mathcal{V}$ -functor $Q : \mathcal{A} \otimes \mathcal{A}^{\text{op}} \otimes \mathcal{A} \longrightarrow \mathcal{A}$ equipped with $\mathcal{V}$ -natural transformations

$$\phi : Q(A, B, Q(C, D, E)) \longrightarrow Q(Q(A, B, C), D, E)$$

$$\xi_A^B : Q(A, B, B) \longrightarrow A$$

$$\zeta_B^A : B \longrightarrow Q(A, A, B) ,$$

such that the following five conditions hold:

$$\begin{array}{ccc}
 Q(Q(A, B, C), D, Q(E, F, G)) & \xleftarrow{\phi} & Q(A, B, Q(C, D, Q(E, F, G))) \\
 \downarrow \phi & & \downarrow Q(1, 1, \phi) \\
 Q(Q(Q(A, B, C), D, E), F, G) & & \\
 \uparrow Q(\phi, 1, 1) & & \\
 Q(Q(A, B, Q(C, D, E)), F, G) & \xleftarrow{\phi} & Q(A, B, Q(Q(C, D, E), F, G)) \\
 \\
 Q(A, B, Q(B, B, C)) & \xrightarrow{\phi} & Q(Q(A, B, B), B, C) \\
 \uparrow Q(1, 1, \zeta) & & \downarrow Q(\xi, 1, 1) \\
 Q(A, B, C) & \xrightarrow{1} & Q(A, B, C)
 \end{array}$$

$$\begin{array}{ccc}
 Q(A, A, Q(B, C, D)) & \xrightarrow{\phi} & Q(Q(A, A, B), C, D) \\
 & \nwarrow \zeta \quad \nearrow Q(\zeta, 1, 1) & \\
 & Q(B, C, D) &
 \end{array}$$

$$\begin{array}{ccc}
 Q(A, B, Q(C, D, D)) & \xrightarrow{\phi} & Q(Q(A, B, C), D, D) \\
 & \searrow Q(1, 1, \xi) \quad \swarrow \xi & \\
 & Q(A, B, C) &
 \end{array}$$

$$\begin{array}{ccc}
 & Q(A, A, A) & \\
 \zeta \nearrow & & \searrow \xi \\
 A & \xrightarrow{1} & A \cdot \square
 \end{array}$$

Flock convolution

- $\mathcal{V}$ -Cocts is the 2-category of cocomplete $\mathcal{V}$ -categories and colimit-preserving (= continuous) $\mathcal{V}$ -functors made monoidal via $\mathcal{X} \boxtimes \mathcal{Y}: \mathcal{V}\text{-Cocts}(\mathcal{X} \boxtimes \mathcal{Y}, \mathcal{Z}) \simeq \mathcal{V}\text{-Sepcocts}(\mathcal{X} \otimes \mathcal{Y}, \mathcal{Z})$.

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- ▶ $\mathcal{V}\text{-Mod}(\mathcal{I}, -) : \mathcal{V}\text{-Mod} \longrightarrow \mathcal{V}\text{-Cocts}$ is a strong monoidal pseudofunctor taking $\mathcal{A}$ to $[\mathcal{A}^{\text{op}}, \mathcal{V}]$.

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- ▶ The basic formula is

$$Q(M, N, L)A = \int^{B, C, D} Q(A; B, C, D) \otimes MB \otimes NC \otimes LD \quad (11)$$

yielding a $\mathcal{V}$ -functor

$$Q : [\mathcal{A}^{\text{op}}, \mathcal{V}] \otimes [\mathcal{A}, \mathcal{V}] \otimes [\mathcal{A}^{\text{op}}, \mathcal{V}] \longrightarrow [\mathcal{A}^{\text{op}}, \mathcal{V}] \quad (12)$$

which is colimit-preserving in each variable separately.

Many closed skew-monoidal categories

According to our previous Proposition, this gives a multitude of examples of closed skew-monoidal $\mathcal{V}$ -categories $[\mathcal{A}^{\text{op}}, \mathcal{V}]$, one for each choice of object K in $\mathcal{A}$ (or its Cauchy completion); namely, we take the tensor product

$$M * N = Q(M, \mathcal{A}(K, -), N)$$

with skew unit $\mathcal{A}(-, K)$.

Abstract quantum categories and bialgebroids

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- ▶ A bialgebroid is *Hopf* when the fusion morphism of the opmonoidal monad is invertible; that is, when the skew flock is a flock.

Bialgebroids

Example

- ▶ Let $\text{Mod}\mathcal{V}$ denote the monoidal full sub-bicategory of $\mathcal{V}\text{-Mod}$ consisting of the one-object $\mathcal{V}$ -categories. So the objects R of $\text{Mod}\mathcal{V}$ are monoids in $\mathcal{V}$.

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- ▶ Hopf bialgebroids are sometimes called “quantum groupoids”.

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- ▶ Quantum categories are bialgebroids C in the bicategory $(\text{Comod}\mathcal{V})^{\text{co}}$ with $j = \varepsilon^*$ where $\mathcal{V}$ is the monoidal category of modules over a division ring k .
- ▶ Hopf quantum categories are *quantum groupoids* as per Chikhladze-Lack-Street (2010).

The Message

$$\frac{\text{Quantum groupoids}}{\text{Flocks}} = \frac{\text{Groups}}{\text{Herds}}$$

The cartesian case

Theorem (Lack-Street)

For a cartesian bicategory $\mathcal{M}$ in the sense of Carboni-Kelly-Walters-Wood (2008), skew monoidales C in $\mathcal{M}$ with unit the unique left adjoint morphism $I \rightarrow C$, amount to monads on C in $\mathcal{M}$.

The bicategory Span of spans of sets can be generalized to a bicategory $\text{Span}(\mathcal{C})$ of spans in a finitely complete category $\mathcal{C}$; taking $\mathcal{C}$ to be the category of sets and functions, we recover Span itself. The bicategory $\text{Span}(\mathcal{C})$ is cartesian and so in there quantum categories are just monads, and, as noted by Bénabou (1967), these are just categories internal to $\mathcal{C}$.

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- ▶ Let $\Delta_{\perp}$ denote the category whose objects are the finite ordinals $\mathbf{n} = \{0, 1, \dots, n-1\}$ with first element (so that $n > 0$), and whose morphisms are first-element-and-order-preserving functions.

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- ▶ $\Delta_{\perp}$ is Hopf left skew monoidal with ordinal sum as tensor product, with unit $\mathbf{1}$, with left unit constraint $\lambda : \mathbf{1} + \mathbf{n} \longrightarrow \mathbf{n}$ the surjection identifying 0 and 1, and with right unit constraint $\rho : \mathbf{m} \longrightarrow \mathbf{m} + \mathbf{1}$ the injection whose image does not contain $m + 1$.

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Theorem (Lack-Street)

The unique strict monoidal functor $\mathbf{Fsk} \longrightarrow \Delta_{\perp}$, which takes the generating object N of $\mathbf{Fsk}$ to the object $\mathbf{1}$ of $\Delta_{\perp}$, is faithful.